DTU

34229/34249 - Cyber Technology/Communication Technologies

Introduction to (Photonic Related) Information Resources

Henning K. Kristensen from DTU Library – hkkr@dtu.dk

Agenda

- **What is the point?**
  - Why to search
  - How to search
- **Tools provided by DTU Library**
  - DTU Findit
  - Web of Science & Scopus
  - Ebook collections and links to more…
- **Short pitch for subject specific tools**
  - IEEE Explorer & SPIE Digital Library
  - Derwent Innovations Index
- **Reference management (Mendeley)**

Questions & discussions are more than welcome as we go along ☺

What is the point?

- We want to conduct a systematic, scientific and comprehensive literature search in all sources relevant to our field of study.
- We want our search to be transparent and reproducerable -> similar to a lab protocol.
- We want to take full advantage of the currated , quality assured, human indexed and expensive bibliographic databases provided by DTU Library (such as Scopus, IEEE Explorer, Derwent II etc..)
- Why are we not using fancy AI or Google Scholar?

Things to consider in the planning phase

- Identify information sources (books, articles (review or original research), reports, patents, facts..)
- Find/locate the information (via library resources, web, friends/colleagues...)
- Get access to information (DTU Findit, other library tools, Google, friends/colleagues..)
- Evaluate the findings (reading snippets/abstracts, choosing, being critical..)
- Manage the findings (export results to reference management, documentation, citations..)

The Matrix

ChatGPT 3.5

You

Dear chat gpt. Can you please help me with synonyms for Software Defined Radio?

ChatGPT

Certainly! Here are some synonyms and related terms for Software Defined Radio (SDR):

1. Software Radio
2. Software-Defined Wireless
3. Cognitive Radio
4. Digital Radio
5. Software-Defined Receiver
6. Flexible Radio
7. Reconfigurable Radio
8. Software-Defined Transceiver
9. Virtual Radio
10. Programmable Radio

You can use AI and/or Wiki when appropriate

Problem Statement: ..something about Medical imaging using Software Defined Radio?

Major Concept		Minor Concept I		Minor Concept II		Minor Concept III
”Software Defined Radio”		”Medical Imaging”		”velocity estimation”		”arotid artery”
OR		**OR**		**OR**		**OR**
SDR	AND	"Imaging Diagnostics"	AND		AND	”blood vessel”
OR		**OR**		**OR**		**OR**
”Cognitive Radio”		Radiology				

The GamePlan

Bibliographic Databases:

- DTU Findit
- Scopus
- Web of Science
- IEEE Explorer*

TextBooks-Clusters:

- ScienceDirect
- SpringerLink
- Wiley Online

Patent Databases:

- Derwent II
- Google Patents

Referencemanagement-tool

E.g.:

- Mendeley
- Bibtex
- Endnote
- Etc.

Publication

E.g.:

- Article for peer-reviewed journal
- Contribution to conference
- Thesis
- Etc.

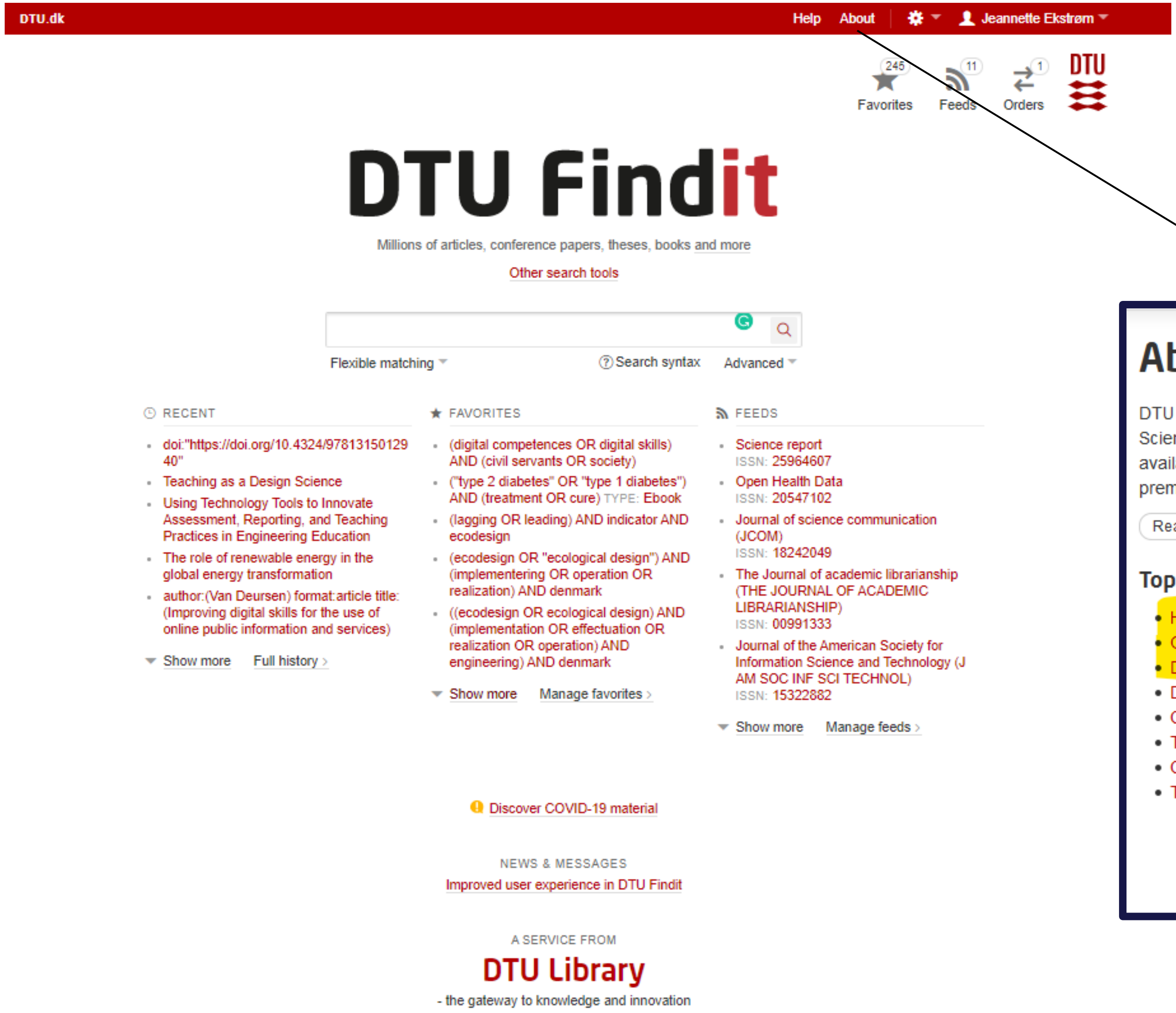
All available resources can be found at: https://findit.dtu.dk/en/about/tools

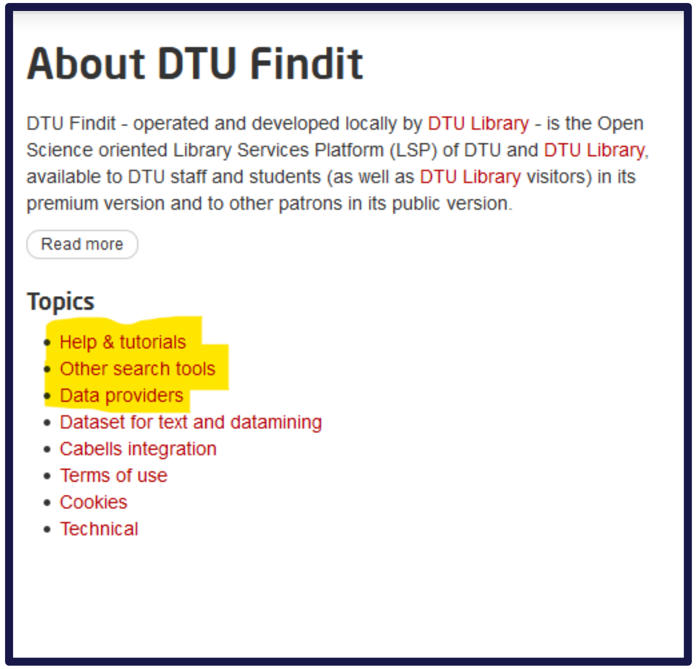
*Journal articles and conference contributions from IEEE are included in Scopus and DTU Findit, but not the IEEE standards.

DTU Findit: the entry point to everything

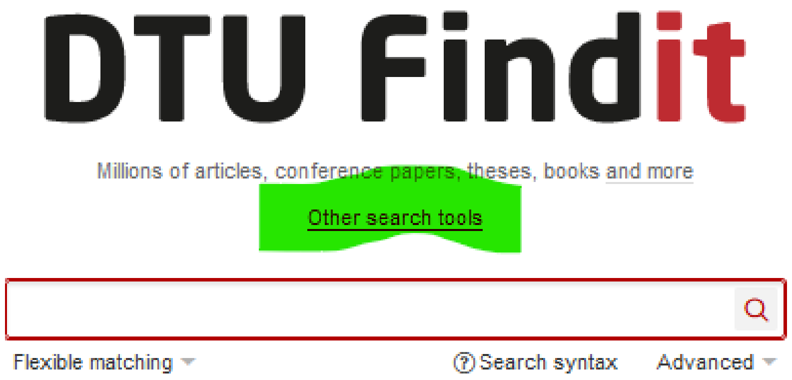
Gives you access to millions of scientific articles, books, theses, reports all through one single entry point –

Mostly in english, but you can search by use of danish keywords (incl. danish articles and book titles)

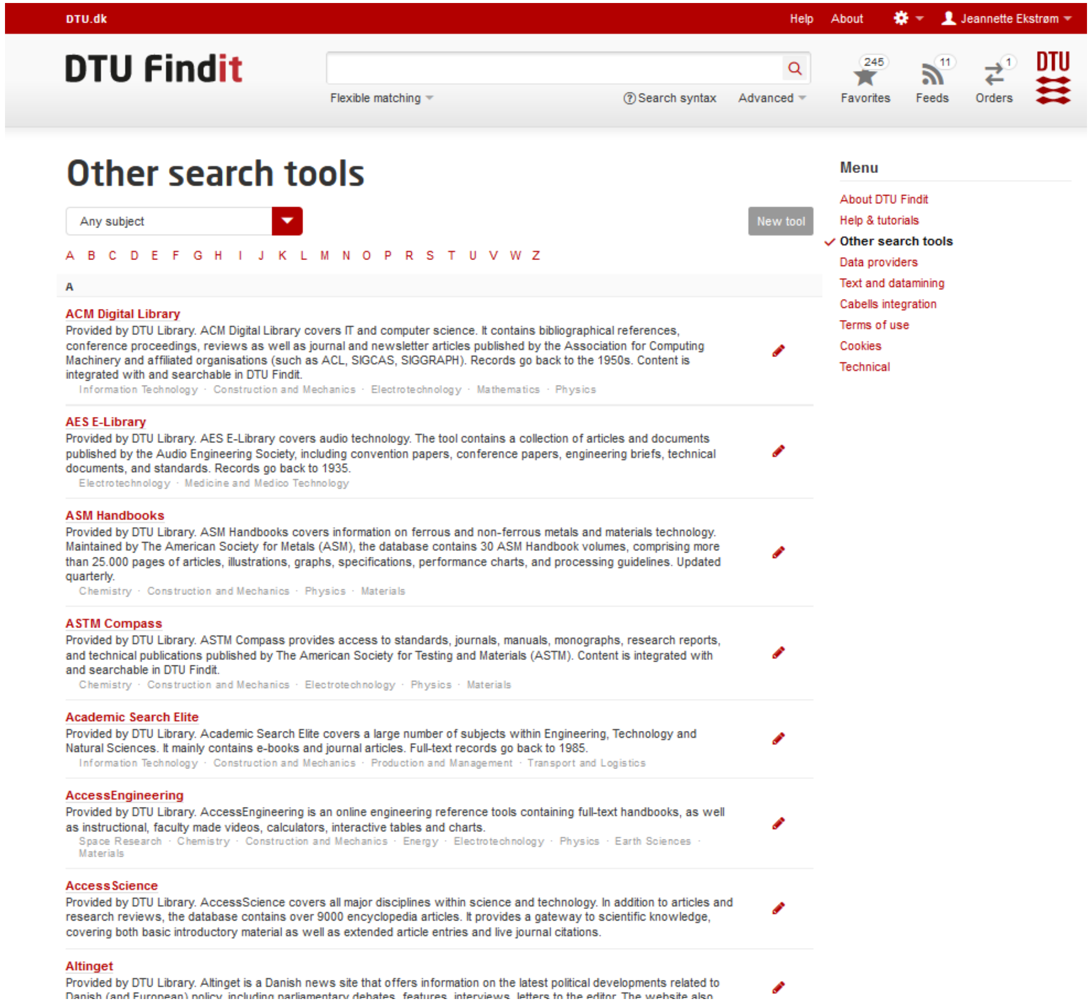




DTU Library provide access to Other Search Tools



https://findit.dtu.dk/en/about/tools



Scopus – bibliographic database. How to use it?

Scopus

Start exploring

Documents | Authors | Researcher Discovery | Organizations

Search tips

Search within
Article title, Abstract, Keywords

Search documents *
"Software Defined Radio" OR sdr OR "Cognitive Radio" ← Koncept I

AND

Search within
Article title, Abstract, Keywords

Search documents
"medical imag*" OR "Imaging Diagnostics" OR Radiology ← Koncept II

+ Add search field | Add date range | Advanced document search >

Reset | Search

Scopus – bibliographic database. How to use it?

Scopus

Search within
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Search documents *
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AND

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Article title, Abstract, Keywords

Search documents
"medical imag*" OR "Imaging Diagnostics" OR radiology

+ Add search field

Reset

Search

Documents | Preprints | Beta Patents | Secondary documents | Research data

88 documents found

Analyze results

All | Export | Download | Citation overview | ••• More

Show all abstracts

Sort by Date (newest)

	Document title	Authors	Source	Year	Citations
1	Conference Paper **Deep Learning based Spectrum predictions for Cognitive Healthcare system**	Kansal, P.	2023 14th International Conference on Computing Communication and Networking Technologies, ICCCNT 2023	2023	0

Available via DTU Findit

Show abstract | DTU Findit | Related documents

Scopus – bibliographic database. How to use it?

IEEE Transactions on Microwave Theory and Techniques • Volume 64, Issue 2, Pages 643 - 652 • 1 February 2016 •
Article number 7378332

Software-defined radar for medical imaging

Marimuthu, Jayaseelan ; Bialkowski, Konstanty S. ;
Abbosh, Amin M.

Abstract

A low-cost reconfigurable microwave transceiver using software-defined radar is proposed for medical imaging. The device, which uses generic software-defined radio (SDR) technology, paves the way to replace the costly and bulky vector network analyzer currently used in the research of microwave-based medical imaging systems. In this paper, calibration techniques are presented to enable the use of SDR technology in a biomedical imaging system. With the aid of an RF circulator, a virtual 1-GHz-wide pulse is generated by coherently adding multiple frequency spectrums together. To verify the proposed system for medical imaging, experiments are conducted using a circular scanning system and directional antenna. The system successfully detects small targets embedded in a liquid emulating the average properties of different human tissues. © 2016 IEEE.

Author keywords

Microwave transceiver; Software-defined radar (SDRadar); Terms-Microwave imaging

Indexed keywords

Engineering controlled terms

Directive antennas; Electric network analyzers; Imaging systems; Microwaves; Radar; Radar imaging; Radio transceivers; Scanning antennas; Software radio; Transceivers

Engineering uncontrolled terms

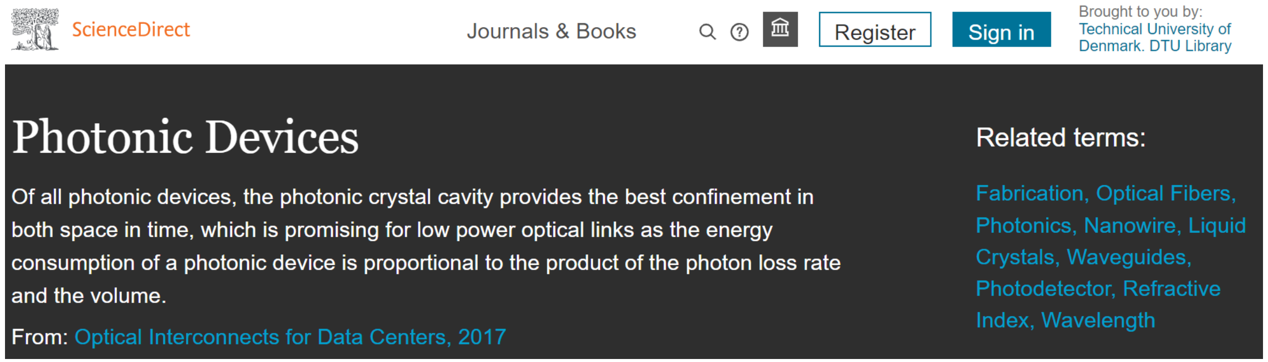
Biomedical imaging; Calibration techniques; Directional Antenna; Microwave imaging; Microwave transceivers; Multiple frequency; Software defined radar; Vector network analyzers

Engineering main heading

Medical imaging

Indexed keywords – the true human value add, that is what we pay for: ←

DTU Library provide access to several E-book collections



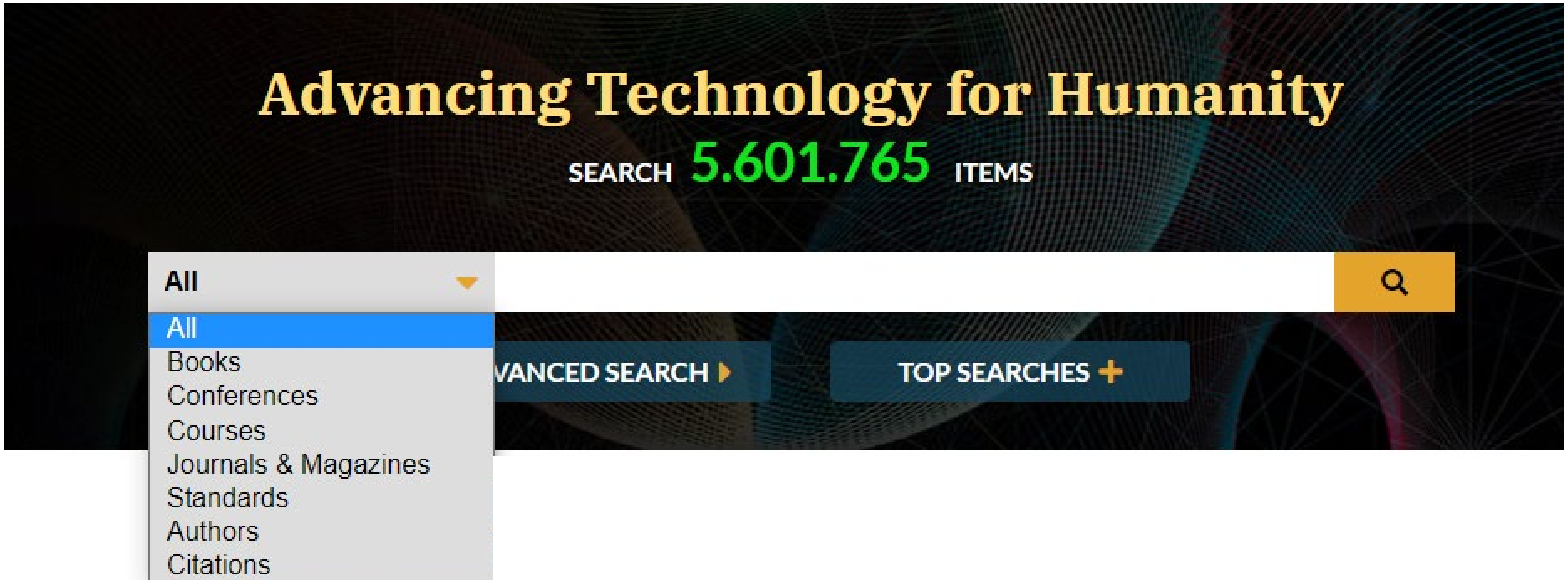
https://www-sciencedirect-com.proxy.findit.cvt.dk/topics/engineering/photonic-devices

Or browse similar subject entries on *'Photonics'*…

Photonics *in Earth and Planetary Sciences*
Photonics *in Engineering*
Photonics *in Physics and Astronomy*
Hamamatsu **Photonics** *in Computer Science*
Microwave **Photonics** *in Physics and Astronomy*
Organic **Photonics** *in Engineering*
Si **Photonics** *in Engineering*
Bio**photonics** *in Engineering*
Bio**photonics** *in Medicine and Dentistry*
Bio**photonics** *in Nursing and Health Professions*
Nano**photonics** *in Physics and Astronomy*

https://link-springer-com.proxy.findit.dtu.dk/

IEEE Xplorer – bibliographic database dedictated to electric engineers



But does it work..?

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IEEE Xplore®

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SEARCH 5.601.765 ITEMS

Authors | henrik | wessing

Showing **1-25** of **29** for **"First Name": henrik** ✖ **"Last Name": wessing** ✖

☐ Conferences (24) ☐ Journals (4) ☐ Magazines (1)

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- Relevance
- Newest First
- Oldest First
- ✔ Most Cited [By Papers]
- Most Cited [By Patents]
- Most Popular
- Publication Title A-Z
- Publication Title Z-A

☐ **Novel scheme for packet forwarding without header modifications in opt**

H. Wessing; H. Christiansen; T. Fjelde; L. Dittmann

Journal of Lightwave Technology

Year: 2002 | Volume: 20, Issue: 8 | Journal Article | Publisher: IEEE

Cited by: Papers (18)

▸ Abstract HTML

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fiber brag |

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Vælg sprog

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PUBLICATION -

- Conference Proceedings (13.018)
  - Paper (12.951)
  - Presentation (785)
  - Poster (46)
- Journal Article (1.135) +
- eBook (91) +

Sort By Relevance Display 25 per page 14.244 results

Proceedings Paper | 20 May 2009 Open Access Paper

Front Matter: Volume 7314

Proc. SPIE. 7314, Photonics in the Transportation Industry: Auto to Aerospace II

Read Abstract +

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Journal Article | 28 August 2014 Open Access

Review of diverse optical fibers used in biomedical research and clinical practice

Gerd Keiser, Fei Xiong, Ying Cui, Perry Ping Shum

Journal of Biomedical Optics Vol. 19, Issue 8 (Aug 2014)

Read Abstract +

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Journal Article | 25 June 2020 Open Access

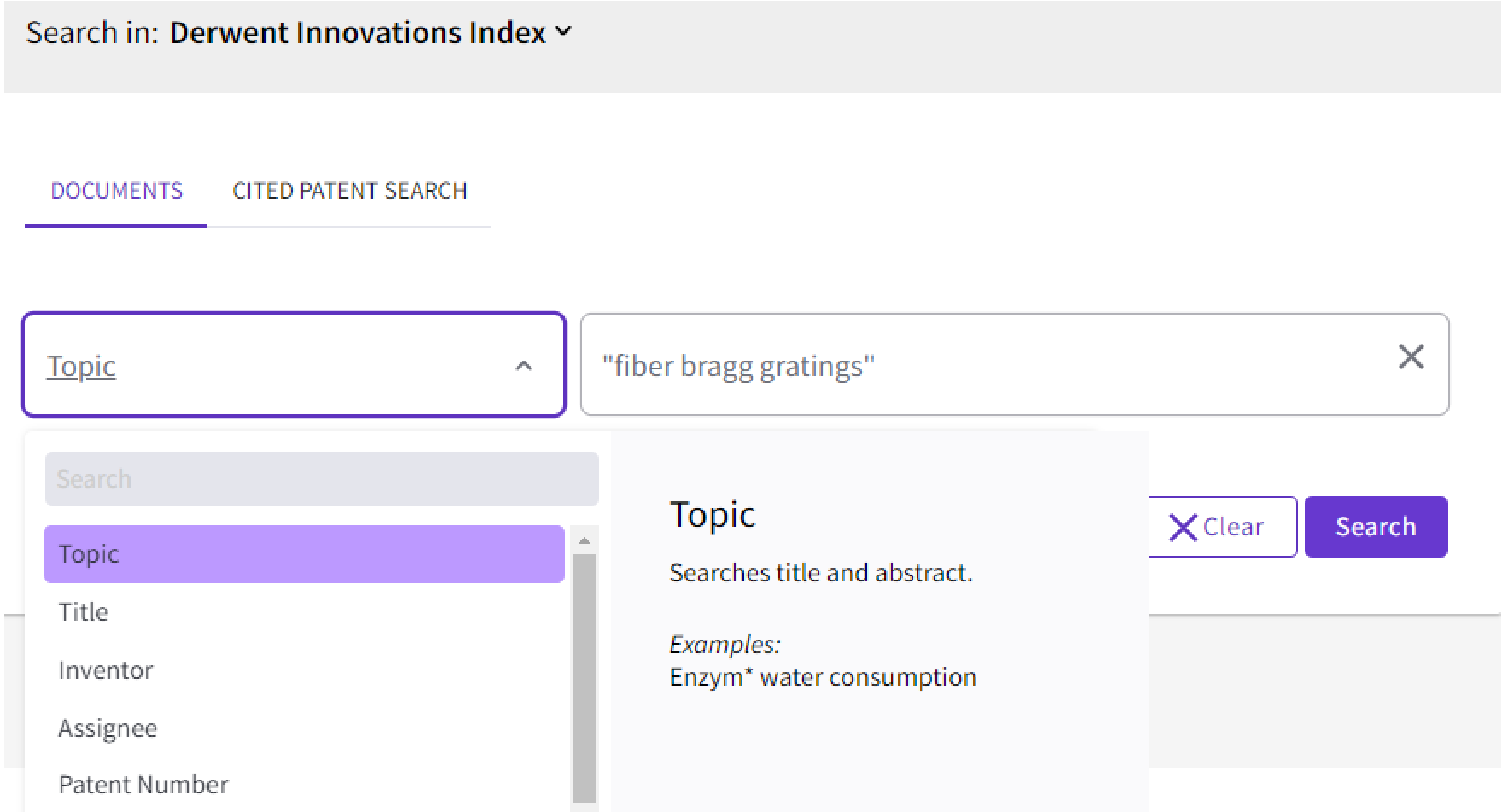
Fiber Bragg grating sensors for monitoring of physical parameters: a comprehensive

PUBLICATION

- Conference Proceedings (13.018)
  - Paper (12.951)
  - Presentation (785)
  - Poster (46)
- Journal Article (1.135) +
- eBook (91) +

Make your information search complete by searching the patent literature

Web of Science™



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623 results from Derwent Innovations Index for:

"fiber bragg gratings" (Topic)

Analyze Results | Create Alert

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Publications | You may also like...

Refine results

Search within results for...

Subject Areas

Engineering	620
Instruments Instrumentation	608
Optics	513
Telecommunications	150
Computer Science	84

See all >

Assignee Names

Baker Hughes Inc	20
Corning Inc	9
Fujikura Ltd	9
Hifi Eng Inc	9
Univ Kunming Sci Technology	9

0/623 Add To Marked List | Export

Sort by: Relevance 1 of 13

1 Optical fiber Raman laser using optical fiber Bragg gratings and remote sensor using the same, particularly for realizing remote sensor for over 25km deep sea and underground

KR2005099744-A; KR669536-B1

Inventor(s) : LEE J H; HAN Y G; (...); KIM S H

Assignee(s) : KOREA INST SCI & TECHNOLOGY

Derwent Primary Accession Number :

2006-666096

3 Citing Patents

2 Parallel optical fiber grating, has four fiber Bragg gratings whose four intercept points are placed on circular cross section of cylinder, where modulation directions of two adjacent fiber Bragg gratings are different by specific degrees

CN109445022-A; CN209265001-U; WO2020140253-A1

Inventor(s) : WANG Y; LIAO C and LI Z

Assignee(s) : UNIV SHENZHEN

Derwent Primary Accession Number :

2019-25197R

The Derwent-record..

Web of Science™

Quantum communication system using code division multiple access network, has three optical fiber elements as basic building blocks of working system, such as fiber Bragg gratings, circulators, and electro-optic modulators

Patent Number: IN202011034133-A

Inventor: SHARMA V

Patent Assignee:
SHARMA V(SHAR-Individual)

Derwent Primary Accession Number: 2022-343496

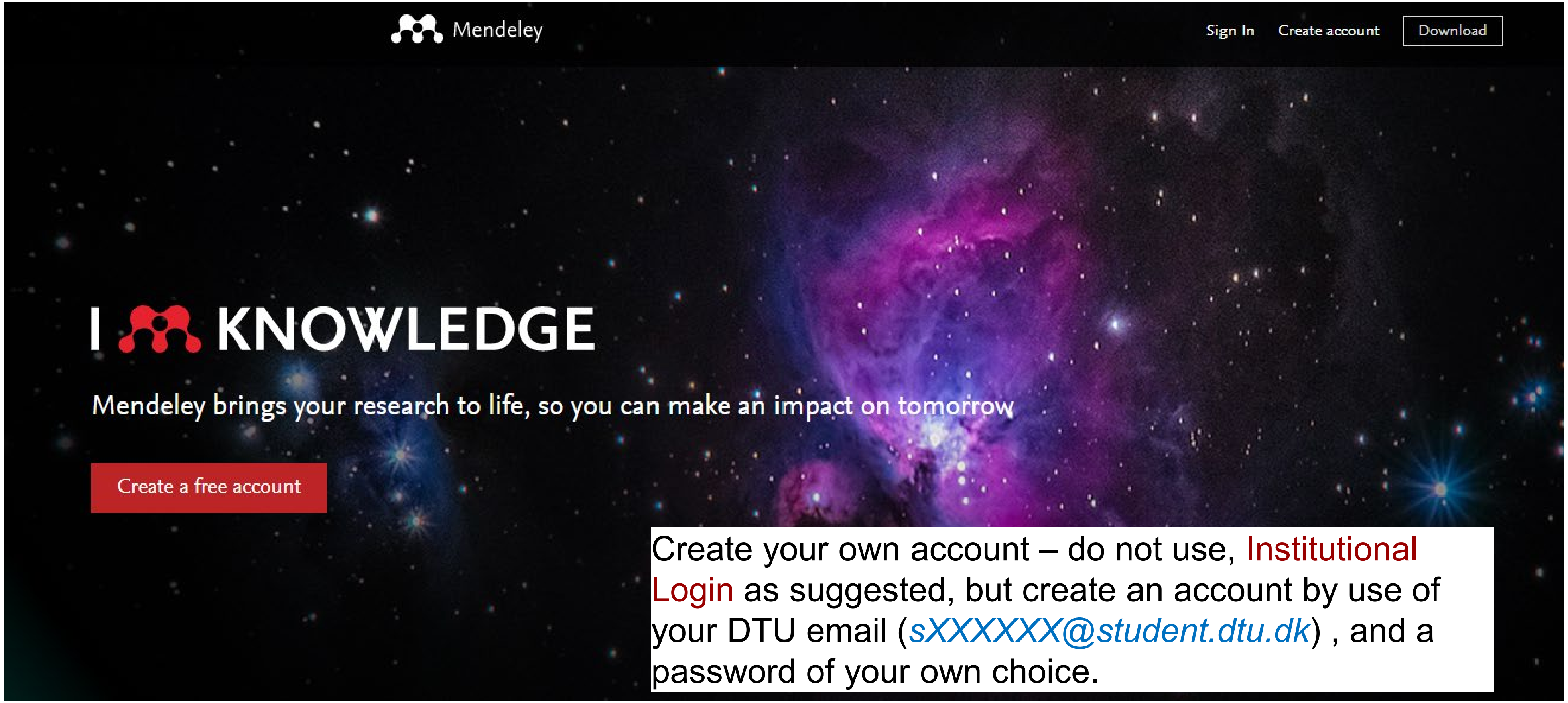
Indexed: 2022-03-27

Abstract:
NOVELTY - The system has three optical fiber units as basic building blocks of a working system. Fiber Bragg gratings (FBG) drop an optical signal by circulators using a dense wavelength division multiplexing (DWDM) technique. The circulators with a center wavelength of 1310 nm, 1550 nm, and 1064 nm are non-reciprocal optical devices that redirect incoming optical signals to successive output ports. The fiber Bragg grating allows multiple of the incoming signals to pass without altering their properties, when a special frequency band is reflected.

USE - Quantum communication system using code division multiple access network for quantum communication protocols such as quantum teleportation, quantum key distribution, and other quantum information networks (Nielsen and Chuang 2001).

ADVANTAGE - The method enables realizing quantum communication with significant reduced information loss using error correction methods, thus enhancing capacity of a quantum channel by correcting errors at a receiver end after prudent encoding in a photon-environment. The method allows concurrent but uncoordinated quantum communication among multiple parties

Introduction to Mendeley & Reference management



Create your own account – do not use, Institutional Login as suggested, but create an account by use of your DTU email (*sXXXXXX@student.dtu.dk*) , and a password of your own choice.

Mendeley is provided by DTU Library

- …….. is a very useful tool when it comes to collecting references, annotate and organise used references, share references and keep track of useful information for Project assignments or other studies

……*this is a very Quick introduction that includes getting access and a few pointers on ”how to get the best out of ..”*

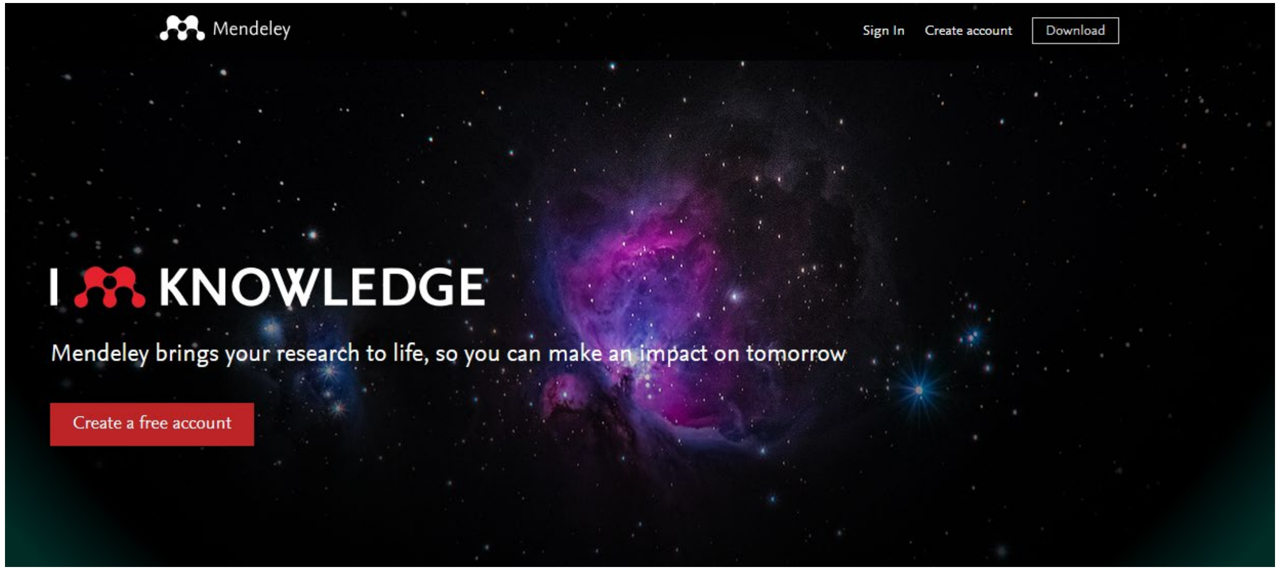
Reference Management by use of Mendeley

Mendeley Institutional Edition via DTU Library
includes a Write'N'Cite plug-in to Word/Libre Office

- Mendeley provides an **empty database, for you to built**
- **Getting started** (create an account at www.mendeley.com)
- **Import references** (from DTU Findit, Web of Science, Scifinder & even Google Scholar)
  - Smart to use the "Mendeley web-importer plug-in" in your browser
- **Citations & Bibliographies are** easily created in Word
  (Mendeley Word Plug-in OR Mendeley Cite app)

See more at the Mendeley help guides - https://www.mendeley.com/guides

Login to www.mendeley.com



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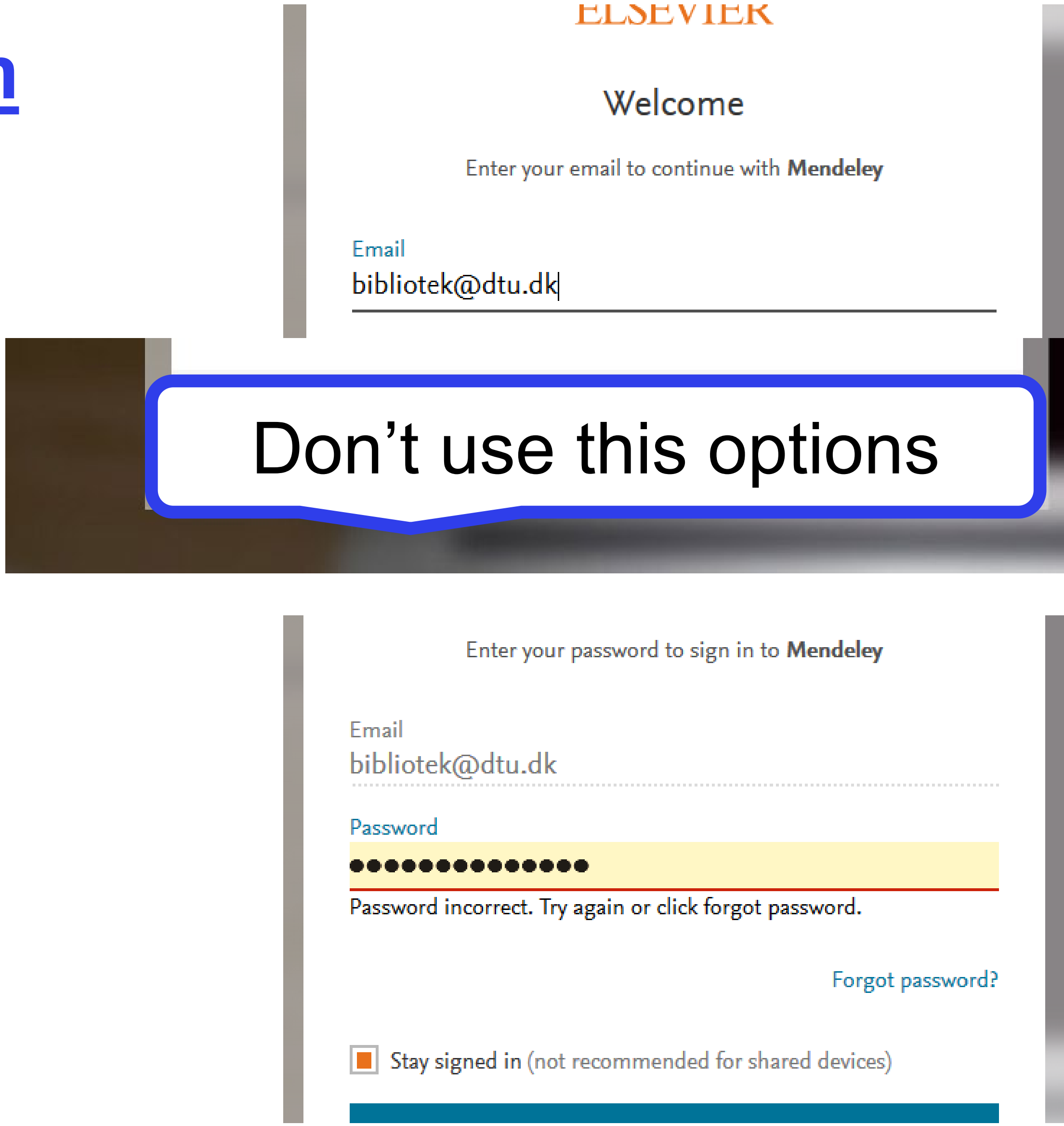


Download Mendeley plug-ins

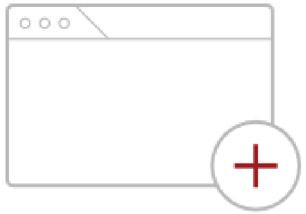
https://www.mendeley.com/download-desktop/

Download Mendeley Desktop for Windows

Download Mendeley Desktop for Windows

Windows 7, 8.1 and 10 (Version 1803) See release notes.

Other systems: Mendeley Desktop for macOS Mendeley Desktop for Linux



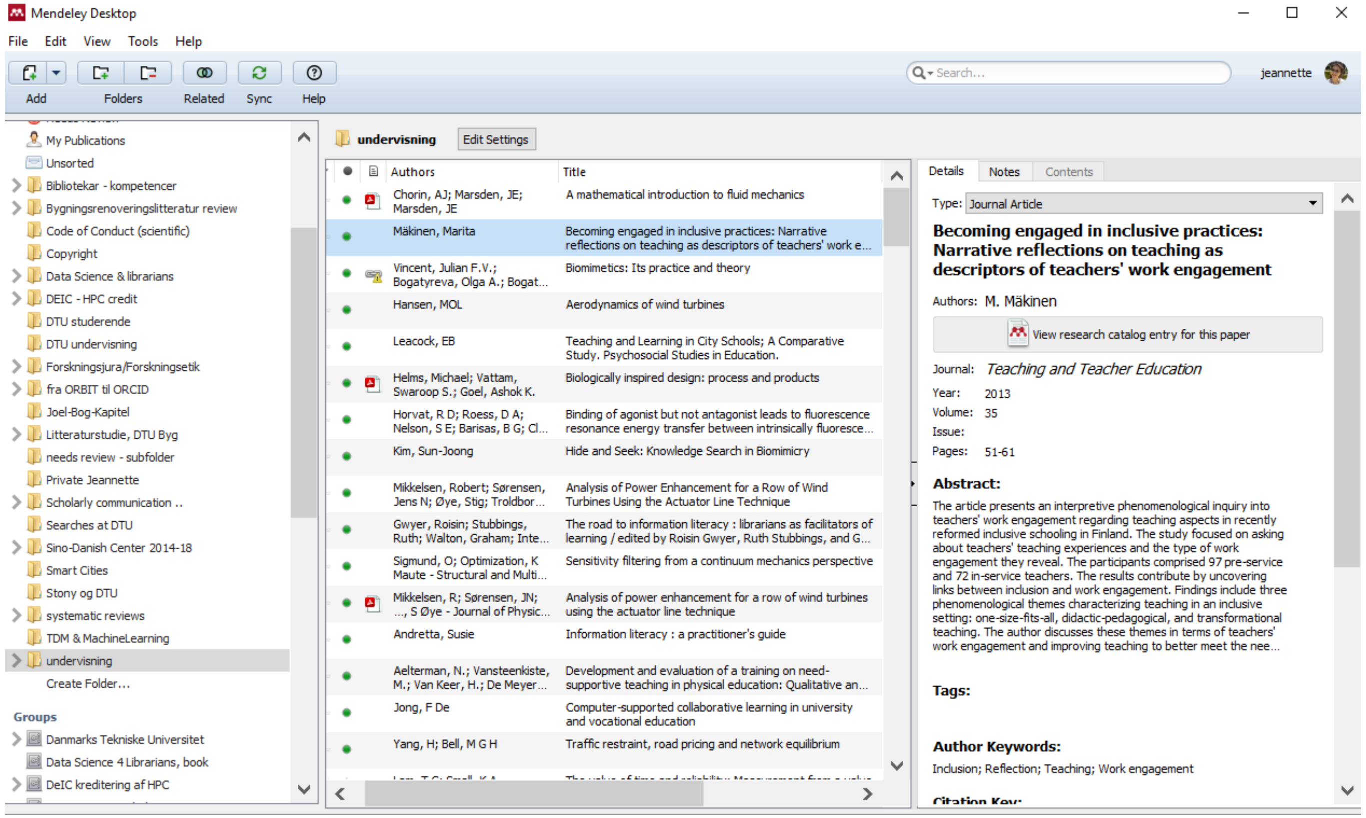
Web Importer

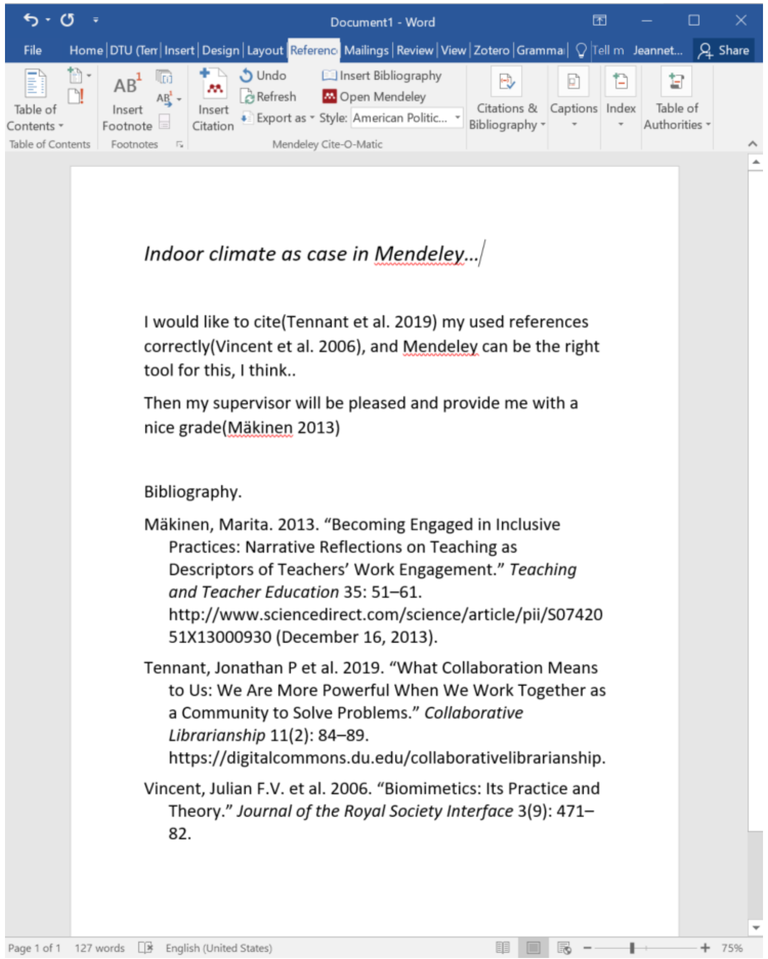
Quickly add articles from any supported website directly into your reference library with a single click using Mendeley's web importer.

Citation Plugin

Mendeley has plug-ins for Word, LibreOffice and BibTeX. We cover thousands of citation styles and you can even create your own.

Working with Mendeley – online, offline, in-text





MAJOR UPCOMING EVENTS

WEBINAR

Building your own library - writing and citing with Mendeley

- reference management with Mendeley

Among other things we can help you with:

- Importing references from DTU Findit, Web of Science, Google Scholar, Scopus, PubMED Reaxys, etc.
- Working with your references in Word or exporting to BibTEX
- How you share references and create private groups
- Tips for using Mendeley for non-Word users (Latex/ Overleaf)

The webinar is just an introduction; we expect that you have made an account with Mendeley beforehand and installed the necessary plugins - see links below.

For more advanced use of Mendeley, please contact us via mail or book an appointment with a librarian.

https://www.bibliotek.dtu.dk/en/calendar/mendeley-20032025?id=b390acf3-4a5a-4984-bb92-7ff1968fae4a

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